

Phase 6: Peripheral Integration - 7-Segment Display and SPI DAC

Objective

Add 7-segment display capability and SPI DAC interface to enable numerical display and analog output for the thermistor monitoring application.

Component BOM

7-Segment Display System

Qty	Component	Function
1	74HC595	8-bit shift register (display driver)
1	Common cathode 7-segment display	Numerical output
8	220Ω resistors	Segment current limiting
1	74F08	AND gates (display control)
1	0.1μF capacitor	Shift register decoupling

SPI DAC System

Qty	Component	Function
1	MCP4921	12-bit DAC with SPI interface
1	74HC14	Schmitt trigger (SPI signal conditioning)
1	10μF capacitor	DAC output filtering
1	0.1μF capacitor	DAC decoupling
3	10kΩ resistors	SPI pull-up resistors

I/O Interface Logic

Qty	Component	Function
1	74LS138	3-to-8 decoder (I/O address decode)
2	74F374N	I/O port latches
1	74LS245	Bidirectional I/O buffer
2	74F08	Control logic gates
4	0.1μF capacitors	Additional decoupling

I/O Memory Map

I/O Address Space (0xA000-0xAFFF)

- 0xA000: 7-segment display data register
- 0xA001: 7-segment display control register
- 0xA010: SPI data register (DAC output)
- 0xA011: SPI control register
- 0xA020: General I/O port A
- 0xA021: General I/O port B
- 0xA030-0xAFFF: Reserved for future peripherals

7-Segment Display Encoding

- Bit Pattern for digits 0-9:
- 0: 0b00111111 (segments a,b,c,d,e,f)
 - 1: 0b00000110 (segments b,c)
 - 2: 0b01011011 (segments a,b,d,e,g)
 - 3: 0b01001111 (segments a,b,c,d,g)
 - 4: 0b01100110 (segments b,c,f,g)
 - 5: 0b01101101 (segments a,c,d,f,g)
 - 6: 0b01111101 (segments a,c,d,e,f,g)
 - 7: 0b00000111 (segments a,b,c)
 - 8: 0b01111111 (segments a,b,c,d,e,f,g)
 - 9: 0b01101111 (segments a,b,c,d,f,g)

Enhanced Instruction Set

I/O Instructions

Opcode	Mnemonic	Description	Encoding
Extended	OUT addr,Rn	Output register to I/O port	Multi-byte
Extended	IN Rn,addr	Input from I/O port to register	Multi-byte
Extended	DISP Rn	Display register value on 7-segment	Single-byte
Extended	DAC Rn	Output register to DAC	Single-byte

Extended Instruction Encoding

I/O instructions use extended opcode format:
Byte 0: 0xFF (Extended instruction prefix)
Byte 1: I/O opcode
Byte 2-3: Address/register parameters (if needed)

Circuit Implementation

7-Segment Display Interface

CPU Data Bus → 74F374N latch → 74HC595 shift register → 7-segment display

Control Sequence:

1. CPU writes display data to I/O address 0xA000
2. Data latched in 74F374N
3. Shift register clocked to update display
4. Segments illuminate according to bit pattern

Segment Mapping:

74HC595 Q0-Q7 → Display segments a,b,c,d,e,f,g,dp

SPI DAC Interface

CPU generates SPI signals:

SCLK: Serial clock (generated by software bit-banging)

MOSI: Data output (12-bit DAC values)

/CS: Chip select (active low)

MCP4921 Configuration:

12-bit DAC with 0-4.096V output range

SPI Mode 0 (CPOL=0, CPHA=0)

16-bit command word format

I/O Address Decoding

Address Bus A15-A12 → 74LS138 decoder

0xA000-0xAFFF generates I/O select signal

Lower address bits select specific I/O registers

Read/Write control manages data direction

Build Steps

1. I/O Address Decoding

- Install 74LS138 decoder for I/O space recognition
- Connect address lines A15-A12 to decoder inputs
- Generate I/O enable signals for 0xA000-0xFFFF range
- Test address decode with logic analyzer

2. 7-Segment Display Driver

- Install 74HC595 shift register
- Connect to 7-segment display with current limiting resistors
- Add control logic for display updates
- Test basic segment illumination

3. Display Data Interface

- Connect CPU data bus to 74F374N display latch
- Wire latch output to 74HC595 shift register input
- Implement display update control logic
- Test numerical display with known bit patterns

4. SPI DAC Implementation

- Install MCP4921 DAC with power supply filtering
- Add 74HC14 Schmitt triggers for SPI signal conditioning
- Implement software SPI bit-banging routines
- Test DAC output with multimeter/oscilloscope

5. I/O Instruction Integration

- Extend instruction decoder for I/O operations
- Add control logic for peripheral access timing
- Implement display and DAC instruction execution
- Test complete I/O instruction set

Test Procedures

1. I/O Address Decode Test

Objective: Verify I/O space recognition and register selection

Manual Test:

1. Set address bus to 0xA000
2. Verify I/O enable signal activates
3. Test address ranges 0xA000, 0xA001, 0xA010, 0xA011
4. Confirm non-I/O addresses don't trigger I/O enable

Expected Results: Logic analyzer shows I/O enable only for 0xA000-0xFFFF

2. 7-Segment Display Test

Objective: Verify display driver functionality

Test Sequence:

1. Manual bit pattern test:
 - Load 0b00111111 → Display should show "0"
 - Load 0b00000110 → Display should show "1"
 - Load 0b01011011 → Display should show "2"

2. Programmatic test:

Address	Data	Expected Display
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0xA000	0x3F	"0"
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0xA000	0x06	"1"
--------	------	-----

0xA000	0x5B	"2"
--------	------	-----

0xA000	0x4F	"3"
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CPU Test Program:

```
0x0000: LD R0,#0x3F ; Load pattern for "0"
0x0002: OUT 0xA000,R0 ; Display "0"
0x0005: LD R0,#0x06 ; Load pattern for "1"
0x0007: OUT 0xA000,R0 ; Display "1"
0x000A: LD R0,#0x5B ; Load pattern for "2"
0x000C: OUT 0xA000,R0 ; Display "2"
```

3. Binary to 7-Segment Conversion Test

Objective: Verify automatic digit conversion

Lookup Table Implementation:

; Store 7-segment patterns in memory

0x8000: 0x3F ; Pattern for digit 0

0x8001: 0x06 ; Pattern for digit 1

0x8002: 0x5B ; Pattern for digit 2

0x8003: 0x4F ; Pattern for digit 3

0x8004: 0x66 ; Pattern for digit 4

0x8005: 0x6D ; Pattern for digit 5

0x8006: 0x7D ; Pattern for digit 6

0x8007: 0x07 ; Pattern for digit 7

0x8008: 0x7F ; Pattern for digit 8

0x8009: 0x6F ; Pattern for digit 9

Conversion Program:

; Display decimal value in R0 (0-9)

0x0000: LD R1,#0x80 ; Load table base address high

0x0002: LD R2,R0 ; Copy digit value

0x0003: ADD R2,R1 ; Calculate table offset

0x0004: LD R3,[R2] ; Load 7-segment pattern

0x0005: OUT 0xA000,R3 ; Display digit

4. SPI DAC Test

Objective: Verify DAC output functionality

MCP4921 Command Format:

Bit 15: Write to DAC register (1)

Bit 14: Buffer control (0 = unbuffered)

Bit 13: Gain select (1 = 1x gain)

Bit 12: Shutdown (1 = active)

Bits 11-0: 12-bit DAC data

SPI Bit-Bang Implementation:

; Send 16-bit word to DAC via SPI

SPI_SEND:

LD R3,#16 ; Bit counter

SPI_LOOP:

; Set MOSI bit based on MSB of data

; Toggle SCLK high then low

; Shift data left

DEC R3

JNZ SPI_LOOP

; Assert /CS to latch data

RET

DAC Test Program:

0x0000: LD R0,#0x30 ; Command: Write, unbuffered, 1x, active

0x0002: LD R1,#0x00 ; MSB of DAC data (0x000 = 0V output)

0x0004: CALL SPI_SEND ; Send to DAC

0x0007: LD R1,#0xFF ; LSB of DAC data (0xFF = max output)

0x0009: CALL SPI_SEND ; Send to DAC

Expected Results:

- 0x3000 command → 0V output
- 0x3FFF command → ~4.096V output

5. Integrated Display/DAC Test

Objective: Demonstrate coordinated peripheral operation

Temperature Simulation Program:

; Simulate temperature display with analog output

0x0000: LD R0,#0x02 ; Temperature = 25°C (display "2")

0x0002: DISP R0 ; Show tens digit

0x0003: LD R0,#0x05 ;

0x0005: DISP R0 ; Show units digit (result: "25")

0x0006: LD R0,#0x66 ; DAC value for 25°C equivalent

0x0008: DAC R0 ; Output analog voltage

0x0009: JMP 0x0000 ; Repeat display

Logic Analyzer Validation

SPI Timing Analysis

Channel	Signal	Purpose
0	SPI Clock (SCLK)	Clock timing verification
1	SPI Data (MOSI)	Data bit sequence
2	Chip Select (/CS)	Frame timing
3-18	SPI Data Word	16-bit command verification
19-26	DAC Output (analog)	Using analog channels

Display Update Timing

Channel	Signal	Purpose
27-34	7-segment data	Display pattern verification
35	Shift register clock	Update timing
36	Latch enable	Display update control
37-44	Display segments	Final output verification

Critical Timing Requirements

- **SPI Setup Time:** Data stable before clock edge (5ns min)
- **SPI Hold Time:** Data stable after clock edge (5ns min)
- **Display Update:** Latch enable pulse width (50ns min)
- **DAC Settling:** Output settling time (4.5µs typical)

Success Criteria

- ☐ I/O address decoding correctly identifies peripheral registers
- ☐ 7-segment display shows correct digits for all patterns
- ☐ Binary to 7-segment conversion works accurately
- ☐ SPI communication transfers data correctly to DAC
- ☐ DAC output voltage matches expected values
- ☐ Display and DAC operations work simultaneously
- ☐ I/O instructions execute within timing requirements
- ☐ No interference between peripheral operations

Performance Validation

Display Update Rate

- Measure maximum display refresh frequency
- Verify flicker-free operation at normal speeds
- Test rapid digit changing sequences

SPI Transfer Speed

- Determine maximum reliable SPI clock rate
- Verify DAC update rate capability
- Test continuous DAC output streaming

System Integration Impact

- Measure CPU performance with peripheral operations
- Verify memory access timing not affected
- Confirm no bus contention during I/O operations

Application Programming Examples

Temperature Display Routine

```
; Convert 8-bit temperature value to dual 7-segment display
TEMP_DISPLAY:
    LD R0,[TEMP_SENSOR] ; Read temperature value
    LD R1,#10           ; Divisor for tens digit
    ; Divide R0 by 10: quotient=tens, remainder=units
    DIV R0,R1           ; Tens in R0, units in R1
    DISP R0             ; Display tens digit
    DISP R1             ; Display units digit
    RET
```

DAC Voltage Output

```
; Convert temperature to proportional voltage
TEMP_TO_VOLTAGE:
    LD R0,[TEMP_SENSOR] ; Read temperature (0-100°C)
    LD R1,#41           ; Scale factor (4095/100)
    MUL R0,R1           ; Scale to 12-bit DAC range
    DAC R0              ; Output voltage (0-4.096V)
    RET
```

Troubleshooting Guide

Issue: 7-Segment Display Shows Wrong Digits

- Verify bit pattern lookup table values
- Check segment wiring (a,b,c,d,e,f,g mapping)
- Confirm current limiting resistor values
- Test shift register clock timing

Issue: SPI DAC Not Responding

- Verify SPI timing with logic analyzer
- Check /CS, SCLK, MOSI connections
- Confirm DAC power supply (2.7V-5.5V)
- Validate 16-bit command format

Issue: Peripheral Interference

- Check I/O address decoding logic
- Verify bus arbitration timing
- Confirm proper tristate control
- Add additional decoupling capacitors

Issue: Timing Violations

- Reduce peripheral access frequency
- Add wait states for slower peripherals
- Verify setup/hold time margins
- Check signal integrity on long traces

Integration Notes for Phase 7

- Document peripheral timing requirements
- Note I/O instruction execution overhead
- Plan thermistor interface requirements
- Record analog output calibration data
- Identify additional peripheral capabilities needed

Documentation Deliverables

- I/O instruction set reference

- Peripheral interface specifications
- SPI protocol implementation details
- 7-segment display driver documentation
- Application programming examples
- Timing analysis and performance data
- Complete peripheral integration schematic